

INTERNSHIP PROJECT REPORT

**DATA CLEANING AND SALES DATA
ANALYSIS USING EXCEL, PYTHON, SQL
AND POWER BI**

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**INTERNSHIP NAME : DATA ANALYST
INTERNSHIP**

ORGANIZATION : CODEORBIT TECH

INTERNSHIP DURATION : 1.5 MONTHS

ACADEMIC YEAR : 2026

OBJECTIVE :

The goal of this project is so clean and analyze sales data using Excel and Python and extract useful business insights from the data. The project also aims to present the results through charts and dashboard, and SQL queries for business insights in a simple and understandable way.

DATASET :

I download the Inventory-Tracking data set from kaggle's website. This dataset is a Excel file.

DATA CLEANING :

In this step data cleaning I used to excel and handled missing values, and corrected the format which all steps are documented and attached this.

PYTHON ANALYSIS :

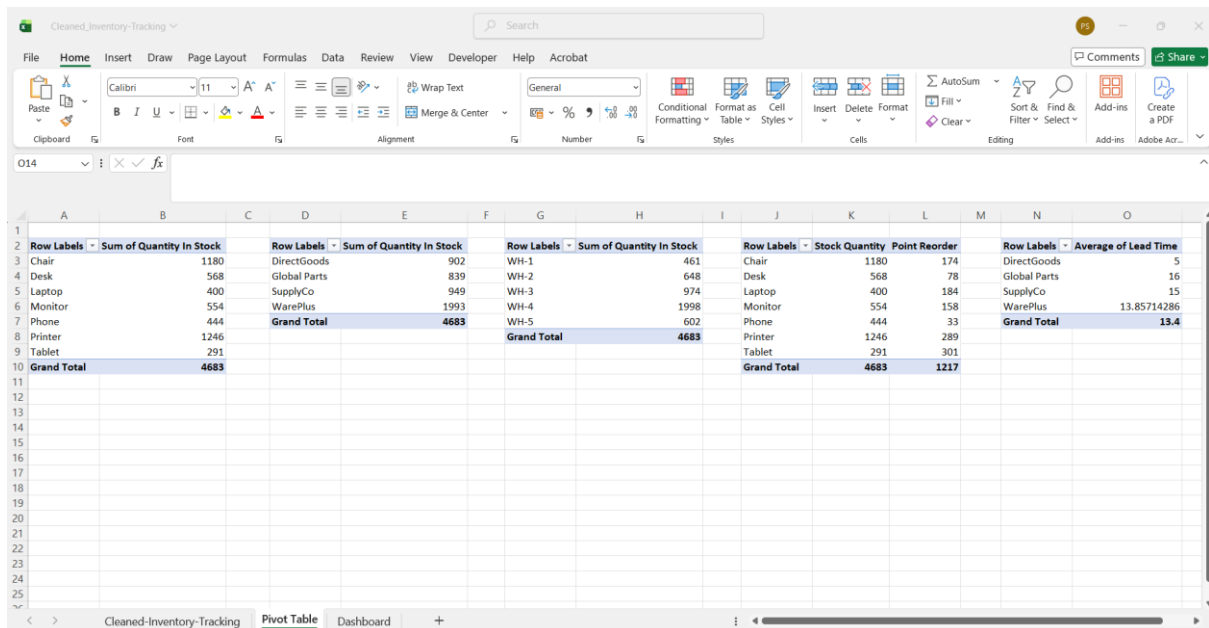
This step I used in Pandas and calculate Number of Unique products, Total Quantity In Stock, Average Quantity In Stock, Product with highest stock, Product with Lowest Stock, Average Unit cost, Number of Products per supplier, Products that need reordering, Inventory value, Total Inventory value , Quantity In Stock by Product, Unit Cost by Product, Stock by Supplier, Stock by Storage Location, Product Distribution, Supplier Distribution, Quantity In Stock Distribution, Lead Time Distribution, Stock vs Reorder Point, Unit Cost Distribution and create bar, pie, histogram, box plot all details are documented and attached this.

DASHBOARD :

I create dashboard in excel using Pivot table. Here the dashboard

- 1.Create Pivot Table
2. All details in Dashboard
- 3.Top Quantity In Stock by Product In Dashboard.

1. Create Pivot Table



Row Labels	Sum of Quantity In Stock
Chair	1180
Desk	568
Laptop	400
Monitor	554
Phone	444
Printer	1246
Tablet	291
Grand Total	4683

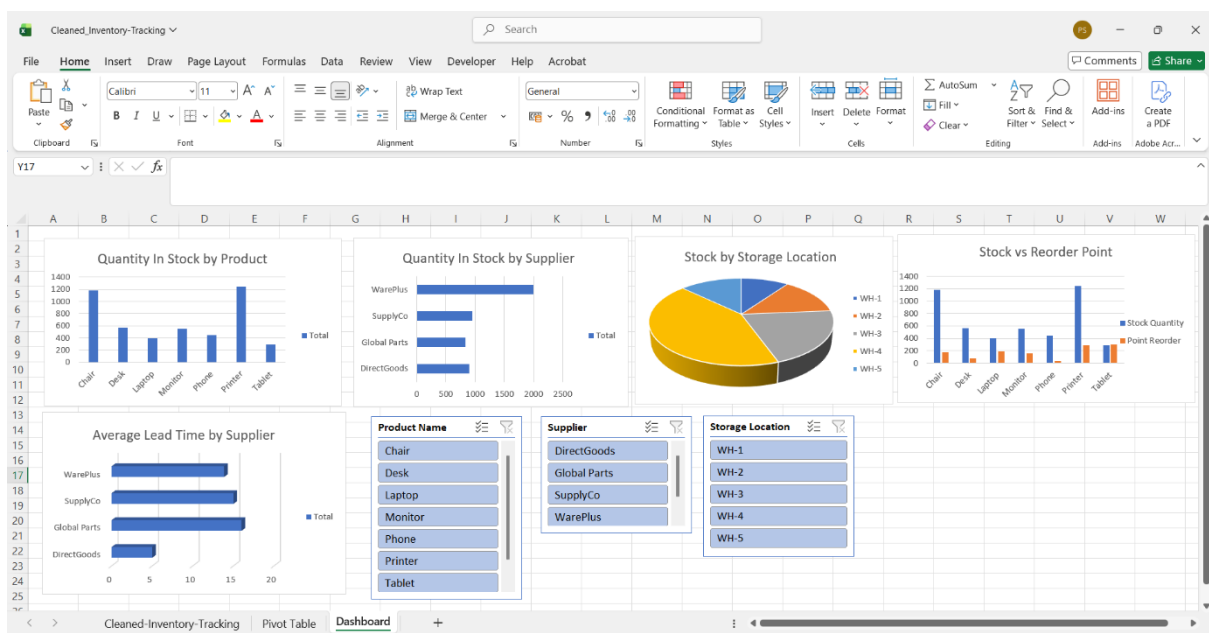
Row Labels	Sum of Quantity In Stock
DirectGoods	902
Global Parts	839
SupplyCo	949
WarePlus	1993
Grand Total	4683

Row Labels	Sum of Quantity In Stock
WH-1	461
WH-2	648
WH-3	974
WH-4	1998
WH-5	602
Grand Total	4683

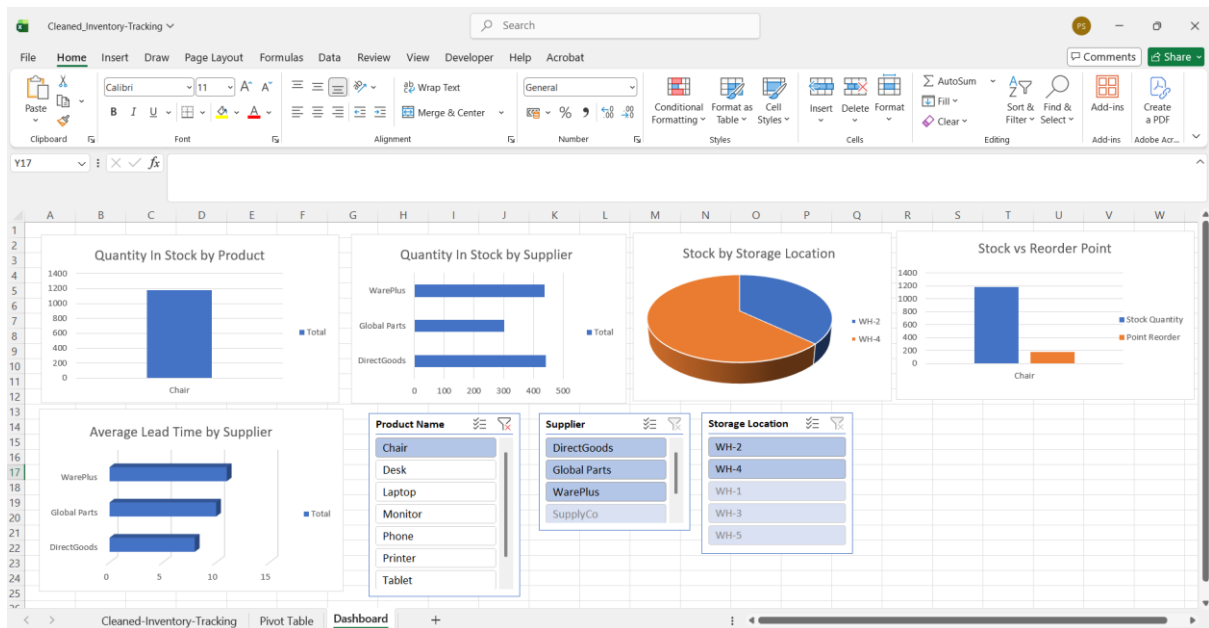
Row Labels	Stock Quantity	Point Reorder
Chair	1180	174
Desk	568	78
Laptop	400	184
Monitor	554	158
Phone	444	33
Printer	1246	289
Tablet	291	301
Grand Total	4683	1217

Row Labels	Average of Lead Time
DirectGoods	5
Global Parts	16
SupplyCo	15
WarePlus	13.85714286
Grand Total	13.4

2. All details in Dashboard



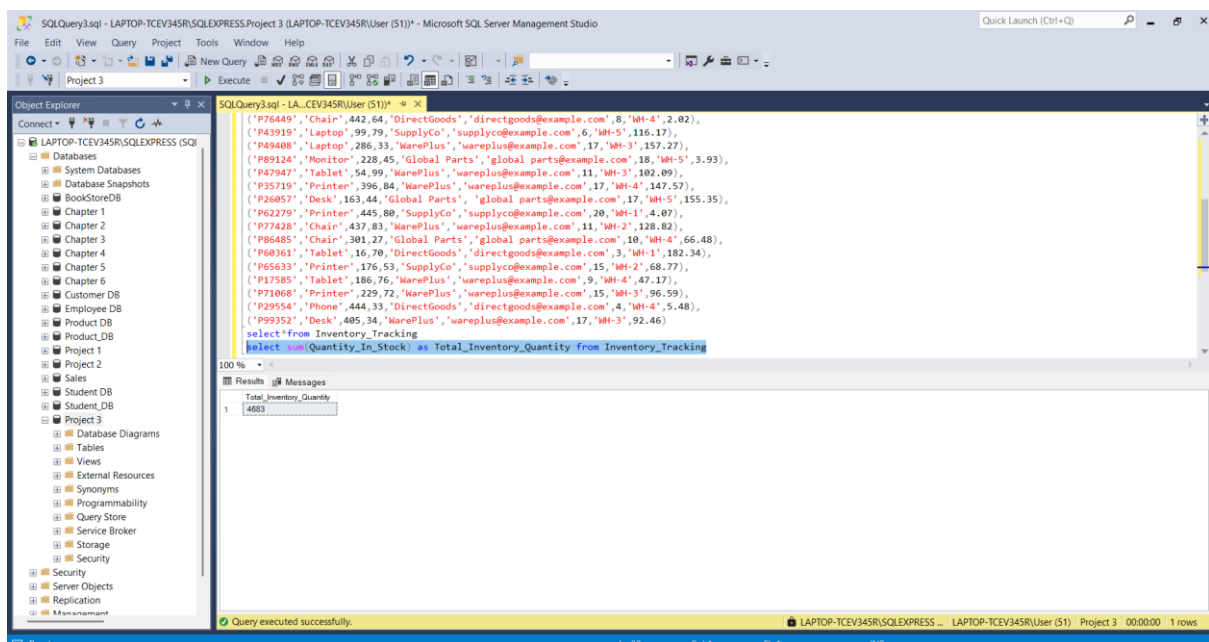
3. Top Quantity In Stock by Product in Dashboard



SQL QUERIES :

In this step I used Sql Server Management Studio and find all the business insights. All steps are documented and attached this. Here the important Queries.

1. Total Inventory Quantity



2. Products below Reorder Point

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Object Explorer' with the 'Databases' folder expanded, showing 'LAPTOP-TCFV345R\SQLEXPRESS (SQL)'. The right pane shows a SQL query window with the following code:

```
SELECT Product_ID, Product_Name, Quantity_In_Stock, Reorder_Point
FROM Inventory_Tracking
WHERE Quantity_In_Stock < Reorder_Point
```

The query results are displayed in a table with 4 rows:

Product_ID	Product_Name	Quantity_In_Stock	Reorder_Point
1	Laptop	15	72
2	Tablet	35	56
3	Tablet	54	99
4	Tablet	16	70

3. Total Quantity by Product

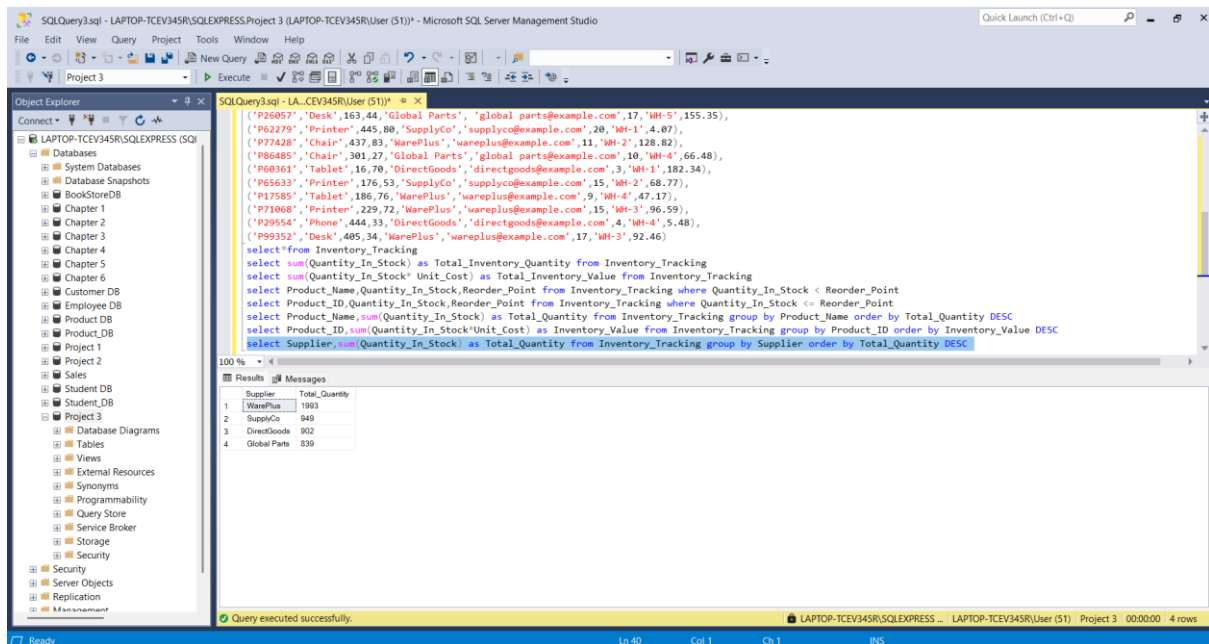
The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Object Explorer' with the 'Databases' folder expanded, showing 'LAPTOP-TCFV345R\SQLEXPRESS (SQL)'. The right pane shows a SQL query window with the following code:

```
SELECT Product_Name, SUM(Quantity_In_Stock) as Total_Quantity
FROM Inventory_Tracking
GROUP BY Product_Name
ORDER BY Total_Quantity DESC
```

The query results are displayed in a table with 7 rows:

Product_Name	Total_Quantity
Printer	1248
Chair	1180
Desk	560
Monitor	554
Phone	444
Laptop	400
Tablet	291

4. Supplier wise Quantity



The screenshot displays the Microsoft SQL Server Management Studio interface. The 'Object Explorer' on the left shows the database structure, including 'Databases', 'Tables', and 'Views'. The 'Query Editor' in the center contains a SQL query that calculates the total inventory quantity for each supplier. The 'Results' pane at the bottom shows the output of the query, which is a table with two columns: 'Supplier' and 'Total_Quantity'.

```
SQLQuery3.sql - LAPTOP-TCEV345R\SQLEXPRESS (User (51)) - Microsoft SQL Server Management Studio
```

```
SELECT * FROM Inventory_Tracking WHERE Product_ID IN ('P26057', 'Desk', 163, 44, 'Global Parts', 'global parts@example.com', 17, 'WH-5', 155.35), ('P62279', 'Printer', 445, 80, 'SupplyCo', 'supplyco@example.com', 20, 'WH-1', 4.07), ('P77428', 'Chair', 437, 83, 'WarePlus', 'wareplus@example.com', 11, 'WH-2', 128.82), ('P86485', 'Chair', 301, 27, 'Global Parts', 'global parts@example.com', 10, 'WH-4', 66.48), ('P60361', 'Tablet', 16, 70, 'DirectGoods', 'directgoods@example.com', 3, 'WH-1', 182.34), ('P65633', 'Printer', 176, 53, 'SupplyCo', 'supplyco@example.com', 15, 'WH-2', 68.77), ('P15585', 'Tablet', 186, 76, 'WarePlus', 'wareplus@example.com', 9, 'WH-4', 47.17), ('P71068', 'Printer', 229, 72, 'WarePlus', 'wareplus@example.com', 15, 'WH-3', 96.59), ('P29554', 'Phone', 444, 33, 'DirectGoods', 'directgoods@example.com', 4, 'WH-4', 5.48), ('P99352', 'Desk', 405, 34, 'WarePlus', 'wareplus@example.com', 17, 'WH-3', 92.46)
```

```
select * from Inventory_Tracking
select sum(Quantity_In_Stock) as Total_Inventory_Quantity from Inventory_Tracking
select sum(Quantity_In_Stock * Unit_Cost) as Total_Inventory_Value from Inventory_Tracking
select Product_Name, Quantity_In_Stock, Reorder_Point from Inventory_Tracking where Quantity_In_Stock < Reorder_Point
select Product_ID, Quantity_In_Stock, Reorder_Point from Inventory_Tracking where Quantity_In_Stock < Reorder_Point
select Product_Name, sum(Quantity_In_Stock) as Total_Quantity from Inventory_Tracking group by Product_Name order by Total_Quantity DESC
select Product_ID, sum(Quantity_In_Stock * Unit_Cost) as Inventory_Value from Inventory_Tracking group by Product_ID order by Inventory_Value DESC
select Supplier, sum(Quantity_In_Stock) as Total_Quantity from Inventory_Tracking group by Supplier order by Total_Quantity DESC
```

Supplier	Total_Quantity
WarePlus	1993
SupplyCo	949
DirectGoods	902
Global Parts	839

Query executed successfully.

CONCLUSION :

This dataset presents various business insights; some of the important queries include which are

- 1.Total Inventory Quantity 4683.
- 2.Products below Reorder Point are Laptop, Tablet, Tablet, Tablet.
- 3.Total Quantity by Product is Printer which is the Highest Total Quantity.
- 4.Supplier wise Quantity is WarePlus which is the Highest Total Quantity.